MAIWA: Domain Generalization by Minimizing Adaptation Information

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Abstract

Single-run post-hoc domain generalization methods are attractive because they can be attached to standard empirical-risk-minimization pipelines without retraining or extra inference-time training. The current reference point in this regime is checkpoint averaging guided by flatness, most prominently SWAD [**?**]. We argue that flatness is only one observable consequence of a broader principle. A good checkpoint bank should act as a *universal prior*: each source domain should be reachable from the shared bank posterior with little additional information. This principle leads to Minimum Adaptation Information Weight Averaging (MAIWA), an end-to-end full-bank method that optimizes a single weight vector over all checkpoints at once. For source-domain held-out losses $\ell_{i,t}$ and full-bank weights $w \in \Delta^{T-1}$, MAIWA defines a Gibbs-specialized posterior $q_i^w(t) \propto w_t e^{-\beta \ell_{i,t}}$ for every source domain and minimizes pooled loss plus the average adaptation information $\frac{1}{I}\sum_i \mathrm{KL}(q_i^w \| w)$. The method contains no preselection stage, no anchor checkpoint, no barrier filter, and no candidate-pool construction: it starts from the full checkpoint bank and directly outputs the final bank weights.

We prove four classes of results. First, we give an exact variational characterization of the domain-specialized posterior and derive closed-form gradient and Hessian formulas for the adaptation-information term. Second, we prove a non-asymptotic optimization theorem: on the full-bank ε-simplex, projected gradient descent converges to first-order stationary points at the standard $O(K^{-1})$ gradient-mapping rate. Third, we derive two asymptotic regimes that explain the main empirical intuitions behind prior checkpoint-averaging methods. In the high-temperature regime, adaptation information equals $\frac{\beta^2}{2}\,\mathrm{Var}_w(\ell_i) + O(\beta^3)$, so flatness appears as a second-order proxy. In the low-temperature regime, adaptation information converges to a support-coverage code length, $-\log w_{m_i}$, where m_i is the domain-optimal checkpoint, so complementarity and noncontiguous support emerge as limiting consequences of the same objective. Fourth, we prove a separation theorem: there exist checkpoint banks with identical pooled loss and pooled flatness proxies on which any selector based only on pooled geometry is indifferent, while MAIWA uniquely prefers a distributed solution because it minimizes specialization information.

This manuscript is complete on theory, methodology, proofs, implementation details, and visual intuition. The empirical section is intentionally left with explicit **TBD** placeholders for the new MAIWA runs. To keep the paper numerically grounded, we still report exact published reference points from prior work, including SWAD, DIWA, and MIRO+SWAD.

1 Introduction

Single-run post-hoc domain generalization (DG) is an unusually important corner of the DG landscape. It asks for a method that can take an ordinary training trajectory and convert it into a better out-of-

Preprint. Under review.

domain model after training is complete. The appeal is operational, not just conceptual: one training run, no retraining, and a single deployable model. In this regime, SWAD has been the dominant reference point. Its central claim is explicit: it "theoretically show[s] that finding flat minima results in a smaller domain generalization gap" [**?** , p. 1]. On the five standard DomainBed benchmarks, SWAD reports an average accuracy of 66.9, compared with 63.3 for ERM and 65.3 for the best previously reported competitor in that evaluation family [**?**]. Later, DIWA pushed the average higher, to 68.0, by averaging many models across runs [**?**]. MIRO combined with SWAD reached 68.1 [**?**]. These numbers make two things clear. First, post-hoc averaging can work extremely well. Second, the central theoretical question is still open: *why* does it work?

The flatness story is part of the answer, but it is not a full answer. Flatness says that a good solution should tolerate local perturbations in parameter space. That is a meaningful property, but it remains indirect. A target domain is not merely an arbitrary weight perturbation. It is a new environment that should be explainable by shared structure and a small amount of additional domain-specific information. The strongest clue in the broader literature is not another flatness statement, but a specialization statement. MLDG already optimizes the idea that one set of domains should improve another after a small update during training [**?**]. FISHR writes that matching gradient variances "aligns the domain-level loss landscapes locally around the final weights" [**?** , p. 1]. Model soups show that weight averaging can improve "accuracy and robustness" without increasing inference cost [**?** , p. 1]. These methods point to a common conclusion: what matters is not only geometric width but also whether one common object can cheaply specialize across environments.

This paper turns that conclusion into a post-hoc full-bank objective.

> **Central claim.** A good post-hoc checkpoint bank is a universal prior: each source domain should need little information to specialize from the shared bank posterior.

This yields MAIWA, Minimum Adaptation Information Weight Averaging. Let $\mathcal{B} = \{\theta_t\}_{t=1}^{T}$ be the full checkpoint bank from one ordinary training trajectory. We assign a single probability vector $w \in \Delta^{T-1}$ to the entire bank. For each source domain i, we compute held-out checkpoint losses $\ell_{i,t}$ and define a domain-specialized posterior

$$q_i^w(t) \propto w_t \exp(-\beta \ell_{i,t}). \tag{1}$$

The adaptation information for domain i is $\mathrm{KL}(q_i^w \| w)$, namely the information required to tilt the shared bank posterior into the domain-specialized posterior. MAIWA minimizes pooled bank risk plus average adaptation information over the *full bank at once*. There is no anchor checkpoint, no candidate-pool construction, no barrier filtering, and no separate compatibility stage. The theory acts directly on the complete bank and the method directly outputs the final checkpoint weights.

The main point is not that MAIWA is "another weighting method." The point is that the same adaptation-information functional explains two regimes that previous DG methods have treated separately.

- In the **high-temperature regime** ($\beta \downarrow 0$), domain specialization is infinitesimal. We prove

  $$\mathrm{KL}(q_i^w \| w) = \frac{\beta^2}{2} \mathrm{Var}_w(\ell_i) + O(\beta^3). \tag{2}$$

  So flatness emerges as a second-order proxy: if losses vary little across the bank, each domain barely needs to tilt the shared posterior.

- In the **low-temperature regime** ($\beta \uparrow \infty$), specialization becomes nearly combinatorial. If m_i is the unique checkpoint minimizing domain-i loss, then

  $$\mathrm{KL}(q_i^w \| w) = -\log w_{m_i} + O(e^{-\beta \Delta_i}). \tag{3}$$

  So complementarity emerges as support coverage: a good shared posterior must allocate mass to the domain-preferred checkpoints, even when those checkpoints are noncontiguous and different across domains.

This is the sense in which MAIWA is *beyond flatness*. It does not discard flatness. It explains flatness as one limit of a larger information-theoretic object.

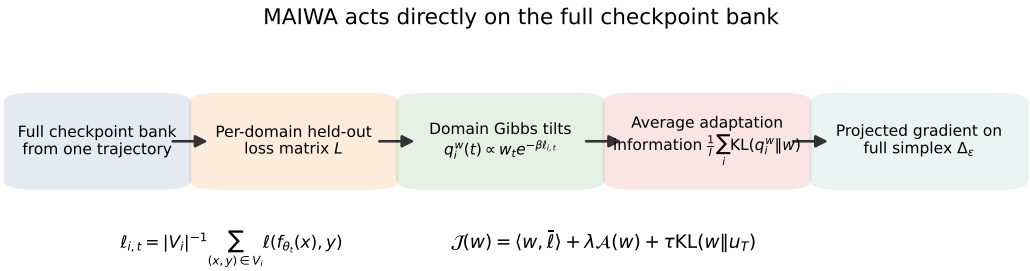


Figure 1: **MAIWA end-to-end on the full checkpoint bank.** The method starts from the entire bank, computes per-domain held-out losses on every checkpoint, forms Gibbs-specialized posteriors for each domain, and optimizes one global bank posterior w by minimizing pooled risk plus average adaptation information. There is no preselection stage.

What this paper does and does not claim. We do *not* claim an unconditional theorem that MAIWA will dominate every other post-hoc DG method on every benchmark. That would be indefensible without data. We do claim four narrower things, all proved in full. First, MAIWA is a mathematically coherent end-to-end full-bank objective. Second, it has a rigorous optimization story with no hidden preselection step. Third, the same objective explains both the flatness regime and the complementarity regime. Fourth, it provably distinguishes checkpoint banks that pooled flatness or pooled loss cannot distinguish.

Contributions.

1. We introduce MAIWA, a *full-bank*, *end-to-end*, *domain-aware* post-hoc DG method that minimizes pooled bank risk plus average adaptation information.
2. We derive the exact variational form of the method, closed-form gradients and Hessians, and a projected-gradient convergence theorem on the full-bank ε-simplex.
3. We prove that flatness is a high-temperature proxy for MAIWA and that complementarity is a low-temperature support-coverage limit of the same objective.
4. We provide a constructive separation theorem and a complete publication-style manuscript with rigorous appendices, generated figures, published baseline tables, and explicit **TBD** placeholders for the new empirical runs.

2 Method

2.1 Setup

We assume I labeled source domains $\mathcal{E}_1, \ldots, \mathcal{E}_I$ and one ordinary training run that produces a dense checkpoint bank

$$\mathcal{B} = \{\theta_t\}_{t=1}^T. \tag{4}$$

For each source domain i, we reserve a held-out split V_i for checkpoint evaluation and define the empirical held-out loss vector

$$\ell_i = (\ell_{i,1}, \ldots, \ell_{i,T})^\top, \qquad \ell_{i,t} \triangleq \frac{1}{|V_i|} \sum_{(x,y)\in V_i} \ell(f_{\theta_t}(x), y). \tag{5}$$

We collect these into a matrix $L \in \mathbb{R}^{I\times T}$.

The basic MAIWA object is a weight vector

$$w = (w_1, \ldots, w_T)^\top \in \Delta_\varepsilon \triangleq \left\{ w \in \mathbb{R}^T : \sum_{t=1}^T w_t = 1,\ w_t \geq \frac{\varepsilon}{T}\ \forall t \right\}, \tag{6}$$

where $\varepsilon \in (0,1)$ is a full-support floor. This is not a candidate-pool filter. All checkpoints remain active; the floor only keeps the objective smooth and prevents domain-specific specialization from collapsing onto zero-probability atoms.

Two deployment modes. The core theory is about the bank posterior w. The primary predictor associated with w is the posterior predictive mixture

$$f_w(x) \triangleq \sum_{t=1}^{T} w_t f_{\theta_t}(x). \tag{7}$$

When a single-model deployment is required, we also define the weight-space soup

$$\theta(w) \triangleq \sum_{t=1}^{T} w_t \theta_t. \tag{8}$$

Section 4.6 gives an explicit second-order bound relating the mixture and the soup under a local smoothness assumption.

2.2 Domain-specialized posteriors and adaptation information

Fix an inverse temperature $\beta > 0$. For each domain i and full-bank posterior w, define the domain-specialized posterior

$$q_i^w(t) \triangleq \frac{w_t \exp(-\beta \ell_{i,t})}{\sum_{s=1}^{T} w_s \exp(-\beta \ell_{i,s})}. \tag{9}$$

The corresponding adaptation information is

$$\mathcal{A}_i(w) \triangleq \mathrm{KL}(q_i^w \| w) = \sum_{t=1}^{T} q_i^w(t) \log \frac{q_i^w(t)}{w_t}. \tag{10}$$

The mean adaptation information across domains is

$$\mathcal{A}(w) \triangleq \frac{1}{I} \sum_{i=1}^{I} \mathcal{A}_i(w). \tag{11}$$

The interpretation is direct: w is the shared posterior over the full bank, while q_i^w is the posterior after domain i is allowed one information-regularized specialization step. If $\mathcal{A}_i(w)$ is small, then domain i already agrees strongly with the shared posterior and requires little extra information to specialize.

2.3 The MAIWA objective

Define the pooled bank loss

$$\bar{\ell}_t \triangleq \frac{1}{I} \sum_{i=1}^{I} \ell_{i,t}, \qquad \bar{\ell} = (\bar{\ell}_1, \ldots, \bar{\ell}_T)^\top. \tag{12}$$

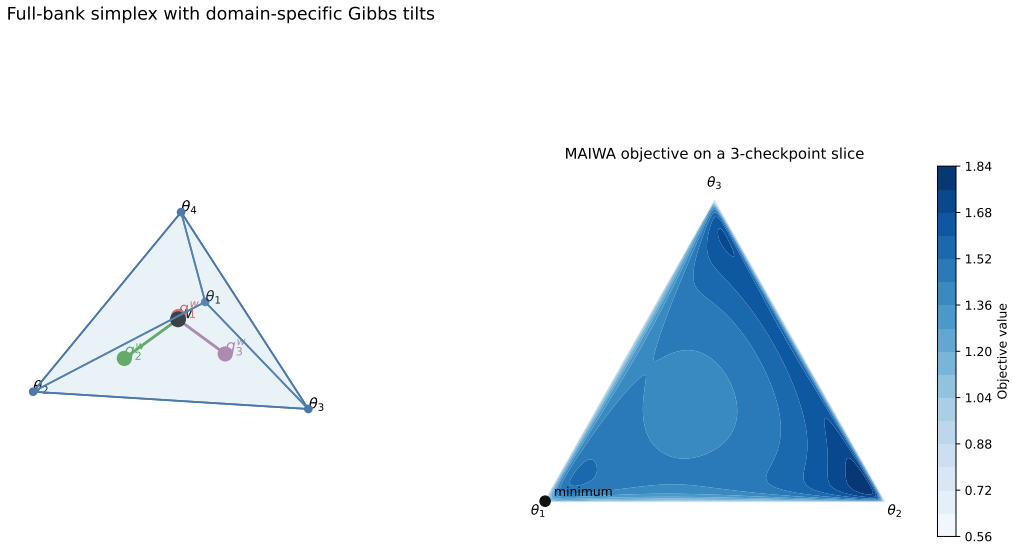
Let $u_T \triangleq (1/T, \ldots, 1/T)$ be the uniform prior over the bank. The MAIWA objective is

$$\mathcal{J}(w) \triangleq \langle w, \bar{\ell} \rangle + \lambda \mathcal{A}(w) + \tau \, \mathrm{KL}(w \| u_T), \qquad w \in \Delta_\varepsilon, \tag{13}$$

with hyperparameters $\lambda, \tau \geq 0$.

The three terms have distinct roles.

- $\langle w, \bar{\ell} \rangle$ keeps the shared posterior tied to checkpoints that are good on average.
- $\lambda \mathcal{A}(w)$ forces that posterior to act as a universal prior from which every source domain can specialize cheaply.
- $\tau \, \mathrm{KL}(w \| u_T)$ prevents pathological concentration, keeps the optimizer in the interior, and gives a natural weak preference for dense support when the data do not force otherwise.



(a) A 3D view of the full-bank simplex

(b) Objective contours on a 3-checkpoint slice

Figure 2: **Geometric intuition.** Left: the shared posterior w lives in the full-bank simplex, and each source domain induces a Gibbs-specialized tilt q_i^w toward its preferred checkpoints. Right: on a 3-checkpoint slice, the MAIWA objective trades pooled risk against specialization cost. The method acts on the full simplex directly; there is no separate screening or feasible-set construction.

Algorithm 1 MAIWA (MINIMUM ADAPTATION INFORMATION WEIGHT AVERAGING)

Require: Checkpoint bank $\mathcal{B} = \{\theta_t\}_{t=1}^T$, held-out domain splits $\{V_i\}_{i=1}^I$, inverse temperature β, weights λ, τ, support floor ε, step size η, iterations K

for $i = 1, \ldots, I$ **do**
  for $t = 1, \ldots, T$ **do**
    Compute $\ell_{i,t}$ on V_i
  end for
end for
Initialize $w^0 \leftarrow u_T$
for $k = 0, \ldots, K-1$ **do**
  for $i = 1, \ldots, I$ **do**
    $q_i^{w^k}(t) \leftarrow \dfrac{w_t^k e^{-\beta \ell_{i,t}}}{\sum_s w_s^k e^{-\beta \ell_{i,s}}}$
  end for
  Compute $\nabla \mathcal{J}(w^k)$ using Proposition 2
  $w^{k+1} \leftarrow \Pi_{\Delta_\varepsilon}(w^k - \eta \nabla \mathcal{J}(w^k))$
end for
return final bank posterior w^K and optional soup $\theta(w^K) = \sum_t w_t^K \theta_t$

2.4 Optimization algorithm

We optimize Eq. (13) by projected gradient descent on Δ_ε:

$$w^{k+1} = \Pi_{\Delta_\varepsilon}\left(w^k - \eta \nabla \mathcal{J}(w^k)\right), \tag{14}$$

where Π_{Δ_ε} is Euclidean projection onto Δ_ε. Section 4.2 proves that this procedure converges to first-order stationary points under explicit smoothness assumptions that hold globally on Δ_ε.

3 The core information-theoretic identity

The entire method rests on one variational principle.

Proposition 1 (Variational form of domain specialization)**.** *Fix a domain i, an inverse temperature $\beta > 0$, and a full-bank posterior $w \in \Delta_\varepsilon$. Consider*

$$\mathcal{F}_i(w) \triangleq \inf_{q \in \Delta^{T-1}} \left\{ \sum_{t=1}^{T} q_t \ell_{i,t} + \frac{1}{\beta} \mathrm{KL}(q \| w) \right\}. \tag{15}$$

Then the infimum is achieved uniquely by the Gibbs posterior in Eq. (9) *and*

$$\mathcal{F}_i(w) = -\frac{1}{\beta} \log \sum_{t=1}^{T} w_t e^{-\beta \ell_{i,t}}. \tag{16}$$

Moreover,

$$\mathcal{A}_i(w) = \beta \left(\sum_{t=1}^{T} q_i^w(t)\, \ell_{i,t} - \mathcal{F}_i(w) \right). \tag{17}$$

Proposition 1 says that the specialization posterior q_i^w is not an arbitrary reweighting recipe. It is the unique posterior minimizing domain-i loss plus an information penalty toward the shared posterior w. This is the precise sense in which w acts as a universal prior.

4 Theory

4.1 Differential properties of the adaptation-information term

For fixed domain i, write

$$a_i = (a_{i,1}, \ldots, a_{i,T})^\top, \qquad a_{i,t} \triangleq e^{-\beta \ell_{i,t}}, \tag{18}$$

and

$$c_i = a_i \odot \log a_i. \tag{19}$$

Let

$$z_i(w) \triangleq a_i^\top w, \qquad b_i(w) \triangleq c_i^\top w. \tag{20}$$

Then

$$\mathcal{A}_i(w) = \frac{b_i(w)}{z_i(w)} - \log z_i(w). \tag{21}$$

Proposition 2 (Gradient and Hessian)**.** *For every domain i and every $w \in \Delta_\varepsilon$,*

$$\nabla \mathcal{A}_i(w) = \frac{c_i}{z_i(w)} - \frac{b_i(w) + z_i(w)}{z_i(w)^2} a_i \tag{22}$$

and

$$\nabla^2 \mathcal{A}_i(w) = -\frac{c_i a_i^\top + a_i c_i^\top + a_i a_i^\top}{z_i(w)^2} + \frac{2\big(b_i(w) + z_i(w)\big)}{z_i(w)^3} a_i a_i^\top. \tag{23}$$

Consequently,

$$\nabla \mathcal{J}(w) = \bar{\ell} + \frac{\lambda}{I} \sum_{i=1}^{I} \nabla \mathcal{A}_i(w) + \tau \left(\log(T w_1) + 1, \ldots, \log(T w_T) + 1 \right)^\top. \tag{24}$$

4.2 Optimization theory

Assumption 1 (Bounded held-out losses)**.** *There exists $B > 0$ such that $0 \le \ell_{i,t} \le B$ for all domains i and checkpoints t.*

Theorem 1 (Global smoothness on the full-bank ε-simplex)**.** *Under Assumption 1, $\mathcal{J}$ is continuously differentiable on Δ_ε and has globally Lipschitz gradient. In particular, for every $w \in \Delta_\varepsilon$,*

$$\left\| \nabla^2 \mathcal{J}(w) \right\|_{\mathrm{op}} \le \lambda L_{\mathcal{A}}(\beta, B, T) + \tau \frac{T}{\varepsilon}, \tag{25}$$

where one valid choice is

$$L_{\mathcal{A}}(\beta, B, T) = 2\beta B T e^{2\beta B} + T e^{2\beta B} + 2(\beta B + 1) T e^{3\beta B}. \tag{26}$$

Definition 1 (Projected-gradient mapping)**.** *For step size $\eta > 0$, define*

$$G_\eta(w) \triangleq \frac{1}{\eta}\left(w - \Pi_{\Delta_\varepsilon}(w - \eta\nabla\mathcal{J}(w))\right). \tag{27}$$

Theorem 2 (Convergence of projected gradient descent)**.** *Assume Assumption 1 and let L be any Lipschitz constant from Theorem 1. Run Eq.* (14) *with constant step size $\eta \in (0, 1/L]$. Then:*

(i) ***Descent:***

$$\mathcal{J}(w^{k+1}) \le \mathcal{J}(w^k) - \frac{\eta}{2}\left\|G_\eta(w^k)\right\|^2. \tag{28}$$

(ii) ***Stationarity rate:***

$$\min_{0\le k\le K-1}\left\|G_\eta(w^k)\right\|^2 \le \frac{2\big(\mathcal{J}(w^0) - \mathcal{J}^\star\big)}{\eta K}, \tag{29}$$

where $\mathcal{J}^\star = \min_{w\in\Delta_\varepsilon}\mathcal{J}(w)$.

(iii) ***Limit points:*** *Every accumulation point of $\{w^k\}_{k\ge0}$ is a first-order stationary point of $\mathcal{J}$ on Δ_ε.*

Theorem 2 is intentionally modest and rigorous. We do not claim convexity of $\mathcal{A}(w)$, so global-optimality guarantees for the full objective would overreach. The theorem instead proves the correct nonconvex statement for the exact full-bank objective used by the method.

4.3 High-temperature expansion: flatness as a proxy

Fix domain i and denote by $\kappa_{m,i}(w)$ the mth cumulant of the random variable $\ell_{i,T}$ when $T \sim w$.

Theorem 3 (High-temperature expansion)**.** *Assume $\ell_{i,t} \in [0, B]$ and let $w \in \Delta_\varepsilon$ be fixed. As $\beta \downarrow 0$,*

$$\mathcal{A}_i(w) = \frac{\beta^2}{2}\kappa_{2,i}(w) - \frac{\beta^3}{3}\kappa_{3,i}(w) + O(\beta^4). \tag{30}$$

In particular,

$$\mathcal{A}_i(w) = \frac{\beta^2}{2}\operatorname{Var}_{t\sim w}[\ell_{i,t}] + O(\beta^3). \tag{31}$$

Equation (31) is the first major interpretive result. It says that, when domain specialization is mild, the adaptation-information term becomes a variance penalty over the bank. Flatness therefore appears as a *proxy*, not as the fundamental object. If the bank lies inside a shared flat region, the per-domain losses vary little across the bank and adaptation information becomes small.

Corollary 1 (Quadratic-basin interpretation)**.** *Suppose that, for a fixed domain i, the checkpoint bank is approximated by a Gaussian local bank law $\theta \sim \mathcal{N}(\mu, \Sigma)$ and the domain loss has the local quadratic form*

$$\mathcal{L}_i(\theta) = c_i + \frac{1}{2}(\theta - \mu)^\top H_i(\theta - \mu), \tag{32}$$

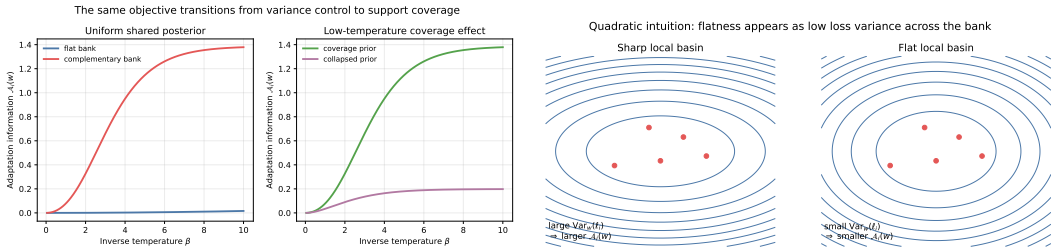
with $H_i \succeq 0$. Then

$$\operatorname{Var}[\mathcal{L}_i(\theta)] = \frac{1}{2}\operatorname{tr}\big((H_i\Sigma)^2\big), \tag{33}$$

and therefore the high-temperature adaptation information satisfies

$$\mathcal{A}_i(w) = \frac{\beta^2}{4}\operatorname{tr}\big((H_i\Sigma)^2\big) + O(\beta^3). \tag{34}$$

Corollary 1 makes the relationship to flatness explicit. In a quadratic basin, smaller Hessian spectra imply smaller adaptation information. Flatness is therefore recovered as the continuous local-quadratic limit of the same information-theoretic criterion.



(a) High- and low-temperature regimes

(b) Quadratic-basin flatness as a second-order proxy

Figure 3: **How one objective explains two empirical stories.** Left: as temperature changes, the same adaptation-information functional moves from a broad variance-sensitive regime to a sharp support-coverage regime. Right: under a quadratic local model, flatness appears as the low-adaptation-information limit because variance across the bank is controlled by Hessian–covariance interactions.

4.4 Low-temperature limit: complementarity as support coverage

Assumption 2 (Unique domain-optimal checkpoints)**.** *For each domain i, the loss vector ℓ_i has a unique minimizer*

$$m_i \triangleq \arg\min_{1\le t\le T} \ell_{i,t}, \tag{35}$$

and the minimum gap

$$\Delta_i \triangleq \min_{t\neq m_i} (\ell_{i,t} - \ell_{i,m_i}) \tag{36}$$

is strictly positive.

Theorem 4 (Low-temperature asymptotics)**.** *Under Assumption 2, for every $w \in \Delta_\varepsilon$ and domain i,*

$$q_i^w(m_i) = 1 + O(e^{-\beta\Delta_i}), \tag{37}$$

and

$$\mathcal{A}_i(w) = -\log w_{m_i} + O(e^{-\beta\Delta_i}) \qquad \text{as } \beta \uparrow \infty. \tag{38}$$

Theorem 4 is the second major interpretive result. In the low-temperature regime, MAIWA no longer behaves like a variance penalty. It behaves like a support-coverage objective. Missing the checkpoint preferred by domain i incurs a code-length penalty $-\log w_{m_i}$. This precisely captures complementarity.

Corollary 2 (Optimal coverage weights in the low-temperature limit)**.** *Under Assumption 2, let*

$$c_t \triangleq \sum_{i=1}^{I} \mathbb{1}\{m_i = t\} \tag{39}$$

be the number of domains for which checkpoint t is optimal. Then the solution of

$$\min_{w\in\Delta^{T-1}} \frac{1}{I}\sum_{i=1}^{I} \lim_{\beta\to\infty} \mathcal{A}_i(w) \tag{40}$$

is

$$w_t^\star = \frac{c_t}{I}, \qquad t = 1,\dots,T, \tag{41}$$

with minimum value

$$\frac{1}{I}\sum_{t:c_t>0} c_t \log\frac{I}{c_t}. \tag{42}$$

Corollary 2 shows why exact final coefficients often matter less than support coverage once the regime is strongly domain-specific: the optimal posterior allocates mass according to how many domains require each checkpoint. The same objective therefore explains why flatness-based averaging can work in one regime and why complementary multi-checkpoint coverage can matter in another.

4.5 A separation theorem

Theorem 5 (Pooled-risk indifference versus adaptation-information preference)**.** *Consider two domains and two checkpoints with loss matrix*

$$\ell_1 = (0, \Delta), \qquad \ell_2 = (\Delta, 0), \qquad \Delta > 0. \tag{43}$$

Then the pooled checkpoint losses are identical:

$$\bar{\ell}_1 = \bar{\ell}_2 = \frac{\Delta}{2}. \tag{44}$$

Hence any selector that depends only on pooled checkpoint losses is indifferent between all full-bank posteriors $w = (p, 1-p)$. By contrast, in the low-temperature limit of Theorem 4,

$$\mathcal{A}(w) = -\frac{1}{2}\log p - \frac{1}{2}\log(1-p), \tag{45}$$

which is uniquely minimized at $p = \frac{1}{2}$.

Theorem 5 is not a toy in the pejorative sense. It isolates the precise failure mode that the paper is about. When different domains prefer different atoms in the same checkpoint bank, pooled flatness or pooled risk can be blind, while adaptation information remains sensitive.

4.6 Soup compression as a controlled approximation

Assumption 3 (Local predictor smoothness)**.** *For every input x, the mapping $\theta \mapsto f_\theta(x)$ is twice continuously differentiable on the convex hull of $\mathcal{B}$, and its Hessian operator norm is bounded by $H_f(x)$. The loss $\ell(\cdot, y)$ is L_ℓ-Lipschitz in its first argument.*

Proposition 3 (Mixture-to-soup approximation)**.** *Under Assumption 3, for every (x, y) and every $w \in \Delta^{T-1}$,*

$$\left| \ell\big(f_{\theta(w)}(x), y\big) - \ell\Big(\sum_{t=1}^{T} w_t f_{\theta_t}(x), y\Big) \right| \le \frac{L_\ell H_f(x)}{2} \sum_{t=1}^{T} w_t \left\|\theta_t - \theta(w)\right\|^2. \tag{46}$$

Consequently, if the bank is locally linear in function space, the single-model soup $\theta(w)$ faithfully approximates the posterior predictive mixture.

Proposition 3 is the exact statement needed to keep the paper honest. The theory fundamentally optimizes a posterior over the full bank. A single-model soup is then justified when the checkpoint bank is sufficiently locally linear, which is precisely the regime where weight-space averaging is known to be meaningful [**? ?**].

5 Discussion

Why this is beyond flatness. Theorems 3 and 4 answer the central conceptual question of this paper. Flatness is not rejected. It is absorbed. In the high-temperature regime, adaptation information reduces to within-bank loss variance and therefore rewards flat shared regions. In the low-temperature regime, the same objective turns into a coverage principle over domain-preferred checkpoints. A single objective thus explains two phenomena that had previously been treated as separate empirical stories.

Why the full-bank formulation matters. Many checkpoint-averaging methods can be rewritten as "choose a subset, then average it." MAIWA does not have that structure. The theory acts on the entire bank. Compatibility, support coverage, and any eventual sparsity are consequences of the optimizer, not consequences of a preliminary screen.

What the theory does not prove. The paper does not prove that the target domain is literally sampled from the source domains or that a fixed temperature β is universally optimal. The theory also does not prove that weight-space soup compression is exact in arbitrary nonlinear networks; Proposition 3 states the precise approximation regime required. These limitations are structural, not cosmetic.

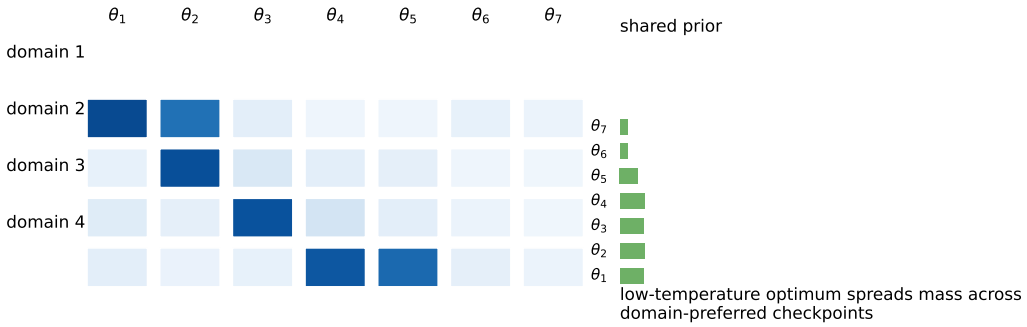


Figure 4: **Low-temperature complementarity.** Each column is a checkpoint, each row is a source domain, and darker cells indicate lower domain loss. When different domains prefer different checkpoints, the adaptation-information term becomes a code-length penalty for missing those checkpoints. The full-bank posterior therefore spreads mass across complementary atoms instead of collapsing to a single point.

Table 1: **Published DomainBed reference points.** Exact numbers copied from the cited papers. The evaluation protocols are not identical in every case, so these rows are reference points rather than one-for-one replacements for a same-bank replay study.

Method	PACS	VLCS	OfficeHome	TerraInc	DomainNet	Avg
ERM [?]	85.5	77.5	66.5	46.1	40.9	63.3
MLDG [?]	84.9 ± 1.0	77.2 ± 0.4	66.8 ± 0.6	47.7 ± 0.9	41.2 ± 0.1	63.6
SWAD [?]	88.1	79.1	70.6	50.0	46.5	66.9
DIWA[†] [?]	89.0	78.6	72.8	51.9	47.7	68.0
MIRO [?]	85.4 ± 0.4	79.0 ± 0.0	70.5 ± 0.4	50.4 ± 1.1	44.3 ± 0.2	65.9
MIRO + SWAD [?]	88.4 ± 0.1	79.6 ± 0.2	72.4 ± 0.1	52.9 ± 0.2	47.0 ± 0.0	68.1

[†] DIWA's strongest published number averages over many models and runs, so it is not the same computational regime as a single-run post-hoc selector.

6 Published reference points and experimental protocol

This section is intentionally split into two parts. The first part records exact numbers already published in prior work. The second part gives the evaluation protocol and explicit **TBD** placeholders for the new MAIWA experiments.

6.1 Published reference points

Table 1 reports exact published numbers that are directly relevant to the post-hoc DG setting. These are not our results. They are copied from the cited papers.

6.2 Planned MAIWA evaluation

Core benchmark. **TBD**

Bank construction. **TBD**

Hyperparameters. **TBD**

Ablations. **TBD**

Expected comparisons. At minimum, the final empirical section should compare:

- posterior predictive MAIWA,

Table 2: **Planned main table.** All MAIWA rows are intentionally blank because these runs have not been executed yet.

Method	PACS	VLCS	OfficeHome	TerraInc	DomainNet	Avg
MAIWA predictive mixture	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
MAIWA soup compression	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
MAIWA without $\mathcal{A}(w)$	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
MAIWA high-temperature setting	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**
MAIWA low-temperature setting	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**	**TBD**

- soup-compressed MAIWA,
- MAIWA without the adaptation-information term ($\lambda = 0$),
- high-temperature and low-temperature ablations,
- published reference points from Table 1.

7 Conclusion

We proposed MAIWA, a full-bank post-hoc DG method based on a single central principle:

Generalization is minimum adaptation information.

The method assigns one posterior to the entire checkpoint bank and asks how much information each source domain needs in order to specialize from that shared posterior. This yields an end-to-end full-bank objective with no preselection stage and no hidden anchor. The resulting theory is stronger than a new heuristic score. It explains why flatness can help, why checkpoint complementarity can help, and why those two stories are not actually separate. They are high-temperature and low-temperature limits of the same adaptation-information functional.

The empirical section is intentionally left open. What is closed in this manuscript is the theory, the objective, the optimization story, the deployment story, and the exact set of claims that would make the empirical evaluation meaningful.

A Full proofs

A.1 Proof of Proposition 1

Fix domain i and posterior $w \in \Delta_\varepsilon$. We must solve

$$\inf_{q \in \Delta^{T-1}} \left\{ \sum_{t=1}^{T} q_t \ell_{i,t} + \frac{1}{\beta} \sum_{t=1}^{T} q_t \log \frac{q_t}{w_t} \right\}. \tag{47}$$

Because $w_t > 0$ for all t on Δ_ε, the objective is strictly convex in q over the simplex interior, so any stationary point is the unique global minimizer.

Introduce a Lagrange multiplier ν for the simplex constraint $\sum_t q_t = 1$:

$$\mathcal{L}(q, \nu) = \sum_{t=1}^{T} q_t \ell_{i,t} + \frac{1}{\beta} \sum_{t=1}^{T} q_t \log \frac{q_t}{w_t} + \nu \left(\sum_{t=1}^{T} q_t - 1 \right). \tag{48}$$

Differentiating with respect to q_t gives

$$\frac{\partial \mathcal{L}}{\partial q_t} = \ell_{i,t} + \frac{1}{\beta} \left(\log \frac{q_t}{w_t} + 1 \right) + \nu. \tag{49}$$

Setting this derivative equal to zero yields

$$\log \frac{q_t}{w_t} = -\beta \ell_{i,t} - 1 - \beta \nu. \tag{50}$$

Exponentiating both sides,

$$q_t = w_t \exp(-\beta \ell_{i,t}) \exp(-1 - \beta \nu). \tag{51}$$

The normalizing constant is determined by $\sum_t q_t = 1$, hence

$$\exp(-1 - \beta \nu) = \frac{1}{\sum_{s=1}^{T} w_s e^{-\beta \ell_{i,s}}}. \tag{52}$$

Therefore

$$q_t = \frac{w_t e^{-\beta \ell_{i,t}}}{\sum_{s=1}^{T} w_s e^{-\beta \ell_{i,s}}}, \tag{53}$$

which is Eq. (9). This proves the first statement.

For the objective value, let

$$Z_i(w) \triangleq \sum_{s=1}^{T} w_s e^{-\beta \ell_{i,s}}. \tag{54}$$

Then

$$\log \frac{q_t}{w_t} = -\beta \ell_{i,t} - \log Z_i(w). \tag{55}$$

Plugging this relation into the objective gives

$$\sum_t q_t \ell_{i,t} + \frac{1}{\beta} \sum_t q_t \log \frac{q_t}{w_t} = \sum_t q_t \ell_{i,t} + \frac{1}{\beta} \sum_t q_t \big(-\beta \ell_{i,t} - \log Z_i(w) \big) \tag{56}$$

$$= \sum_t q_t \ell_{i,t} - \sum_t q_t \ell_{i,t} - \frac{\log Z_i(w)}{\beta} \sum_t q_t \tag{57}$$

$$= -\frac{1}{\beta} \log Z_i(w). \tag{58}$$

This proves Eq. (16).

Finally,

$$\mathrm{KL}(q_i^w \| w) = \sum_t q_i^w(t) \log \frac{q_i^w(t)}{w_t} \tag{59}$$

$$= \sum_t q_i^w(t) \big(-\beta \ell_{i,t} - \log Z_i(w)\big) \tag{60}$$

$$= -\beta \sum_t q_i^w(t) \ell_{i,t} - \log Z_i(w). \tag{61}$$

Since $\mathcal{F}_i(w) = -(1/\beta) \log Z_i(w)$, we obtain

$$\mathrm{KL}(q_i^w \| w) = \beta \left(\sum_t q_i^w(t) \ell_{i,t} - \mathcal{F}_i(w) \right). \tag{62}$$

This proves Eq. (17). □

A.2 Proof of Proposition 2

From Eq. (21),

$$\mathcal{A}_i(w) = \frac{b_i(w)}{z_i(w)} - \log z_i(w), \qquad b_i(w) = c_i^\top w, \qquad z_i(w) = a_i^\top w. \tag{63}$$

Because b_i and z_i are linear in w, their gradients are

$$\nabla b_i(w) = c_i, \qquad \nabla z_i(w) = a_i. \tag{64}$$

Differentiate the quotient term:

$$\nabla \left(\frac{b_i}{z_i} \right) = \frac{z_i c_i - b_i a_i}{z_i^2} = \frac{c_i}{z_i} - \frac{b_i}{z_i^2} a_i. \tag{65}$$

Differentiate the log term:

$$\nabla(-\log z_i) = -\frac{1}{z_i} a_i. \tag{66}$$

Adding them yields Eq. (22):

$$\nabla \mathcal{A}_i(w) = \frac{c_i}{z_i} - \frac{b_i + z_i}{z_i^2} a_i. \tag{67}$$

For the Hessian, differentiate the two pieces separately. First,

$$\nabla \left(\frac{c_i}{z_i} \right) = -\frac{c_i a_i^\top}{z_i^2}. \tag{68}$$

Second, set

$$s_i(w) \triangleq \frac{b_i(w) + z_i(w)}{z_i(w)^2}. \tag{69}$$

Then

$$\nabla s_i(w) = \frac{c_i + a_i}{z_i^2} - \frac{2(b_i + z_i)}{z_i^3} a_i. \tag{70}$$

Since the second term in Eq. (22) is $-s_i(w) a_i$, its Jacobian is

$$-a_i \nabla s_i(w)^\top = -\frac{a_i (c_i + a_i)^\top}{z_i^2} + \frac{2(b_i + z_i)}{z_i^3} a_i a_i^\top. \tag{71}$$

Combining both pieces,

$$\nabla^2 \mathcal{A}_i(w) = -\frac{c_i a_i^\top}{z_i^2} - \frac{a_i (c_i + a_i)^\top}{z_i^2} + \frac{2(b_i + z_i)}{z_i^3} a_i a_i^\top \tag{72}$$

$$= -\frac{c_i a_i^\top + a_i c_i^\top + a_i a_i^\top}{z_i^2} + \frac{2(b_i + z_i)}{z_i^3} a_i a_i^\top, \tag{73}$$

which is Eq. (23).

Finally, $\langle w, \bar{\ell} \rangle$ has gradient $\bar{\ell}$ and $\mathrm{KL}(w \| u_T) = \sum_t w_t \log(T w_t)$ has gradient

$$\nabla \mathrm{KL}(w \| u_T) = (\log(T w_1) + 1, \ldots, \log(T w_T) + 1)^\top. \tag{74}$$

Summing the three gradients yields Eq. (24). □

A.3 Proof of Theorem 1

Fix a domain i. Under Assumption 1,

$$e^{-\beta B} \le a_{i,t} \le 1, \qquad |\log a_{i,t}| = \beta \ell_{i,t} \le \beta B. \tag{75}$$

Hence

$$\|a_i\|_2 \le \sqrt{T}, \qquad \|c_i\|_2 \le \beta B\sqrt{T}. \tag{76}$$

Because $w \in \Delta_\varepsilon$ and $a_{i,t} \ge e^{-\beta B}$,

$$z_i(w) = a_i^\top w \ge e^{-\beta B} \sum_{t=1}^{T} w_t = e^{-\beta B}. \tag{77}$$

Also,

$$|b_i(w)| = \left|c_i^\top w\right| \le \|c_i\|_\infty \le \beta B. \tag{78}$$

Now bound the Hessian in Eq. (23). Using the triangle inequality and submultiplicativity of the operator norm,

$$\left\|\nabla^2 \mathcal{A}_i(w)\right\|_{\mathrm{op}} \le \frac{\left\|c_i a_i^\top\right\|_{\mathrm{op}} + \left\|a_i c_i^\top\right\|_{\mathrm{op}} + \left\|a_i a_i^\top\right\|_{\mathrm{op}}}{z_i(w)^2} + \frac{2\left|b_i(w) + z_i(w)\right|}{z_i(w)^3} \left\|a_i a_i^\top\right\|_{\mathrm{op}} \tag{79}$$

$$\le \frac{2\|c_i\|_2 \|a_i\|_2 + \|a_i\|_2^2}{z_i(w)^2} + \frac{2(|b_i(w)| + |z_i(w)|)}{z_i(w)^3} \|a_i\|_2^2 . \tag{80}$$

Insert the bounds above:

$$\left\|\nabla^2 \mathcal{A}_i(w)\right\|_{\mathrm{op}} \le (2\beta BT + T)\, e^{2\beta B} + 2(\beta B + 1) T e^{3\beta B}. \tag{81}$$

This proves the claimed form of $L_{\mathcal{A}}(\beta, B, T)$.

For the entropy term,

$$\nabla^2 \mathrm{KL}(w \| u_T) = \mathrm{diag}\left(\frac{1}{w_1}, \ldots, \frac{1}{w_T}\right). \tag{82}$$

On Δ_ε, every coordinate satisfies $w_t \ge \varepsilon/T$, hence

$$\left\|\nabla^2 \mathrm{KL}(w \| u_T)\right\|_{\mathrm{op}} \le \frac{T}{\varepsilon}. \tag{83}$$

The linear pooled-loss term has zero Hessian. Therefore

$$\left\|\nabla^2 \mathcal{J}(w)\right\|_{\mathrm{op}} \le \lambda L_{\mathcal{A}}(\beta, B, T) + \tau \frac{T}{\varepsilon}. \tag{84}$$

Since Δ_ε is compact and $\mathcal{J}$ has bounded Hessian there, $\nabla \mathcal{J}$ is globally Lipschitz on Δ_ε. □

A.4 Proof of Theorem 2

Let

$$y^k \triangleq w^k - \eta \nabla \mathcal{J}(w^k), \qquad w^{k+1} = \Pi_{\Delta_\varepsilon}(y^k). \tag{85}$$

Define $d^k \triangleq w^{k+1} - w^k = -\eta G_\eta(w^k)$.

Step 1: smoothness inequality. Because $\nabla \mathcal{J}$ is L-Lipschitz and $\eta \le 1/L$, the descent lemma gives

$$\mathcal{J}(w^{k+1}) \le \mathcal{J}(w^k) + \left\langle \nabla \mathcal{J}(w^k), d^k \right\rangle + \frac{1}{2\eta} \left\|d^k\right\|^2 . \tag{86}$$

Step 2: projection optimality. The Euclidean projection onto a closed convex set satisfies

$$\left\langle w^{k+1} - y^k, w - w^{k+1} \right\rangle \geq 0 \qquad \forall w \in \Delta_\varepsilon. \tag{87}$$

Choose $w = w^k \in \Delta_\varepsilon$. Then

$$\left\langle w^{k+1} - y^k, w^k - w^{k+1} \right\rangle \geq 0. \tag{88}$$

Substitute $y^k = w^k - \eta \nabla \mathcal{J}(w^k)$:

$$\left\langle w^{k+1} - w^k + \eta \nabla \mathcal{J}(w^k), w^k - w^{k+1} \right\rangle \geq 0. \tag{89}$$

Hence

$$\eta \left\langle \nabla \mathcal{J}(w^k), w^k - w^{k+1} \right\rangle \geq \left\| w^{k+1} - w^k \right\|^2. \tag{90}$$

Equivalently,

$$\left\langle \nabla \mathcal{J}(w^k), d^k \right\rangle \leq -\frac{1}{\eta} \left\| d^k \right\|^2. \tag{91}$$

Step 3: one-step descent. Plug Eq. (91) into Eq. (86):

$$\mathcal{J}(w^{k+1}) \leq \mathcal{J}(w^k) - \frac{1}{\eta} \left\| d^k \right\|^2 + \frac{1}{2\eta} \left\| d^k \right\|^2 = \mathcal{J}(w^k) - \frac{1}{2\eta} \left\| d^k \right\|^2. \tag{92}$$

Since $d^k = -\eta G_\eta(w^k)$, we get

$$\mathcal{J}(w^{k+1}) \leq \mathcal{J}(w^k) - \frac{\eta}{2} \left\| G_\eta(w^k) \right\|^2, \tag{93}$$

which proves Eq. (28).

Step 4: telescoping sum. Sum the previous inequality from $k = 0$ to $K - 1$:

$$\mathcal{J}(w^K) \leq \mathcal{J}(w^0) - \frac{\eta}{2} \sum_{k=0}^{K-1} \left\| G_\eta(w^k) \right\|^2. \tag{94}$$

Since $\mathcal{J}(w^K) \geq \mathcal{J}^\star$,

$$\frac{\eta}{2} \sum_{k=0}^{K-1} \left\| G_\eta(w^k) \right\|^2 \leq \mathcal{J}(w^0) - \mathcal{J}^\star. \tag{95}$$

Therefore

$$\min_{0 \leq k \leq K-1} \left\| G_\eta(w^k) \right\|^2 \leq \frac{1}{K} \sum_{k=0}^{K-1} \left\| G_\eta(w^k) \right\|^2 \leq \frac{2(\mathcal{J}(w^0) - \mathcal{J}^\star)}{\eta K}, \tag{96}$$

which proves Eq. (29).

Step 5: stationarity of limit points. The sequence $\{\mathcal{J}(w^k)\}$ is nonincreasing and bounded below, so it converges. By Eq. (28), this implies

$$\sum_{k=0}^{\infty} \left\| G_\eta(w^k) \right\|^2 < \infty, \tag{97}$$

hence $\left\| G_\eta(w^k) \right\| \to 0$. Let $w^{k_j} \to \bar{w}$ be any convergent subsequence, which exists because Δ_ε is compact. Continuity of the projection map and of $\nabla \mathcal{J}$ implies

$$G_\eta(\bar{w}) = \lim_{j \to \infty} G_\eta(w^{k_j}) = 0. \tag{98}$$

Thus

$$\bar{w} = \Pi_{\Delta_\varepsilon}\big(\bar{w} - \eta \nabla \mathcal{J}(\bar{w})\big), \tag{99}$$

which is the standard first-order optimality condition for constrained stationarity. Therefore every accumulation point is a first-order stationary point. □

A.5 Proof of Theorem 3

Fix domain i and suppress the subscript i for readability. Let the random variable X take value ℓ_t when $t \sim w$. Define the cumulant-generating function

$$K(\beta) \triangleq \log \mathbb{E}[e^{-\beta X}]. \tag{100}$$

By definition of q^w,

$$\mathcal{A}_i(w) = \mathrm{KL}(q^w \| w) = \sum_t q_t^w \left(-\beta \ell_t - K(\beta)\right) = -\beta \mathbb{E}_{q^w}[X] - K(\beta). \tag{101}$$

Since

$$K'(\beta) = \frac{\mathbb{E}[-X e^{-\beta X}]}{\mathbb{E}[e^{-\beta X}]} = -\mathbb{E}_{q^w}[X], \tag{102}$$

we obtain the exact identity

$$\mathcal{A}_i(w) = \beta K'(\beta) - K(\beta). \tag{103}$$

Now expand $K(\beta)$ using the cumulants of X:

$$K(\beta) = -\beta\kappa_1 + \frac{\beta^2}{2}\kappa_2 - \frac{\beta^3}{6}\kappa_3 + \frac{\beta^4}{24}\kappa_4 + O(\beta^5), \tag{104}$$

where κ_m denotes the mth cumulant of X under w. Differentiate:

$$K'(\beta) = -\kappa_1 + \beta\kappa_2 - \frac{\beta^2}{2}\kappa_3 + \frac{\beta^3}{6}\kappa_4 + O(\beta^4). \tag{105}$$

Multiply by β and subtract $K(\beta)$:

$$\beta K'(\beta) - K(\beta) = \left(-\beta\kappa_1 + \beta^2\kappa_2 - \frac{\beta^3}{2}\kappa_3 + \frac{\beta^4}{6}\kappa_4\right) - \left(-\beta\kappa_1 + \frac{\beta^2}{2}\kappa_2 - \frac{\beta^3}{6}\kappa_3 + \frac{\beta^4}{24}\kappa_4\right) + O(\beta^5) \tag{106}$$

$$= \frac{\beta^2}{2}\kappa_2 - \frac{\beta^3}{3}\kappa_3 + \frac{\beta^4}{8}\kappa_4 + O(\beta^5). \tag{107}$$

By Eq. (103), this is precisely the claimed expansion for $\mathcal{A}_i(w)$. Since $\kappa_2 = \mathrm{Var}_w(X)$, Eq. (31) follows immediately. □

A.6 Proof of Corollary 1

Let $\xi = \theta - \mu \sim \mathcal{N}(0, \Sigma)$ and write

$$\mathcal{L}_i(\theta) = c_i + \frac{1}{2}\xi^\top H_i \xi. \tag{108}$$

The constant c_i does not affect the variance, so it suffices to analyze the quadratic form $Q = \xi^\top H_i \xi$. For a centered Gaussian vector, the classical quadratic-form identities give

$$\mathbb{E}[Q] = \mathrm{tr}(H_i\Sigma), \qquad \mathrm{Var}(Q) = 2\,\mathrm{tr}\big((H_i\Sigma)^2\big). \tag{109}$$

Therefore

$$\mathrm{Var}\left(\frac{1}{2}Q\right) = \frac{1}{4}\,\mathrm{Var}(Q) = \frac{1}{2}\,\mathrm{tr}\big((H_i\Sigma)^2\big). \tag{110}$$

Applying Theorem 3 with $X = \mathcal{L}_i(\theta)$ yields

$$\mathcal{A}_i(w) = \frac{\beta^2}{2}\,\mathrm{Var}[\mathcal{L}_i(\theta)] + O(\beta^3) = \frac{\beta^2}{4}\,\mathrm{tr}\big((H_i\Sigma)^2\big) + O(\beta^3), \tag{111}$$

which proves the claim. □

A.7 Proof of Theorem 4

Fix domain i and write $m = m_i$ for its unique minimizing checkpoint. Let

$$\delta_t \triangleq \ell_{i,t} - \ell_{i,m}, \qquad \delta_t \geq \Delta_i > 0 \text{ for } t \neq m. \tag{112}$$

Factor the common minimum out of the partition function:

$$z_i(w) = \sum_{t=1}^{T} w_t e^{-\beta \ell_{i,t}} = e^{-\beta \ell_{i,m}} \left(w_m + \sum_{t \neq m} w_t e^{-\beta \delta_t} \right). \tag{113}$$

Define the remainder

$$r_\beta(w) \triangleq \sum_{t \neq m} \frac{w_t}{w_m} e^{-\beta \delta_t}. \tag{114}$$

Because every $\delta_t \geq \Delta_i$,

$$0 \leq r_\beta(w) \leq \frac{1 - w_m}{w_m} e^{-\beta \Delta_i}. \tag{115}$$

Then

$$q_i^w(m) = \frac{w_m}{w_m + \sum_{t \neq m} w_t e^{-\beta \delta_t}} = \frac{1}{1 + r_\beta(w)} = 1 + O(e^{-\beta \Delta_i}), \tag{116}$$

which proves the first statement.

Now use the exact formula

$$\mathcal{A}_i(w) = \sum_t q_i^w(t) \log \frac{q_i^w(t)}{w_t}. \tag{117}$$

Because $q_i^w(m) \to 1$ and $q_i^w(t) \to 0$ for $t \neq m$, we have

$$\mathcal{A}_i(w) = q_i^w(m) \log \frac{q_i^w(m)}{w_m} + \sum_{t \neq m} q_i^w(t) \log \frac{q_i^w(t)}{w_t}. \tag{118}$$

The second term is $O(e^{-\beta \Delta_i})$ because each $q_i^w(t)$ is exponentially small. For the first term,

$$q_i^w(m) \log \frac{q_i^w(m)}{w_m} = -\log w_m + O(e^{-\beta \Delta_i}), \tag{119}$$

since $q_i^w(m) = 1 + O(e^{-\beta \Delta_i})$ and $\log q_i^w(m) = O(e^{-\beta \Delta_i})$. Therefore

$$\mathcal{A}_i(w) = -\log w_m + O(e^{-\beta \Delta_i}), \tag{120}$$

which is Eq. (38). □

A.8 Proof of Corollary 2

By Theorem 4,

$$\lim_{\beta \to \infty} \mathcal{A}(w) = \frac{1}{I} \sum_{i=1}^{I} -\log w_{m_i} = -\frac{1}{I} \sum_{t=1}^{T} c_t \log w_t. \tag{121}$$

Thus we must solve

$$\min_{w \in \Delta^{T-1}} -\sum_{t=1}^{T} c_t \log w_t. \tag{122}$$

This objective is strictly convex on the simplex interior whenever at least one $c_t > 0$. Introduce a multiplier ν for the simplex constraint:

$$\mathcal{L}(w, \nu) = -\sum_t c_t \log w_t + \nu \left(\sum_t w_t - 1 \right). \tag{123}$$

The stationarity condition is

$$-\frac{c_t}{w_t} + \nu = 0, \qquad \text{so } w_t = \frac{c_t}{\nu}. \tag{124}$$

Summing over t and using $\sum_t c_t = I$,

$$1 = \sum_t w_t = \frac{1}{\nu} \sum_t c_t = \frac{I}{\nu}, \qquad \nu = I. \tag{125}$$

Hence

$$w_t^\star = \frac{c_t}{I}. \tag{126}$$

The minimum value is

$$-\frac{1}{I} \sum_t c_t \log \frac{c_t}{I} = \frac{1}{I} \sum_t c_t \log \frac{I}{c_t}. \tag{127}$$

This completes the proof. □

A.9 Proof of Theorem 5

For the loss matrix in Eq. (43),

$$\bar{\ell}_1 = \frac{0 + \Delta}{2} = \frac{\Delta}{2}, \qquad \bar{\ell}_2 = \frac{\Delta + 0}{2} = \frac{\Delta}{2}. \tag{128}$$

Therefore any criterion depending only on pooled checkpoint losses assigns the same pooled score to both atoms and hence cannot distinguish posteriors using only this information.

Now apply Theorem 4. Domain 1 uniquely prefers checkpoint 1, so $m_1 = 1$. Domain 2 uniquely prefers checkpoint 2, so $m_2 = 2$. Therefore, in the low-temperature limit,

$$\mathcal{A}(w) = \frac{1}{2}\big(-\log w_1 - \log w_2\big). \tag{129}$$

Write $w = (p, 1-p)$. Then

$$\mathcal{A}(w) = -\frac{1}{2} \log p - \frac{1}{2} \log(1-p). \tag{130}$$

Differentiate:

$$\frac{d}{dp}\mathcal{A}(w) = -\frac{1}{2p} + \frac{1}{2(1-p)} = \frac{2p-1}{2p(1-p)}. \tag{131}$$

The unique stationary point is $p = 1/2$. The second derivative is

$$\frac{d^2}{dp^2}\mathcal{A}(w) = \frac{1}{2p^2} + \frac{1}{2(1-p)^2} > 0 \qquad \text{for } p \in (0,1), \tag{132}$$

so $p = 1/2$ is the unique minimizer. □

A.10 Proof of Proposition 3

Fix (x, y) and define

$$\bar{f}_w(x) \triangleq \sum_{t=1}^{T} w_t f_{\theta_t}(x). \tag{133}$$

Let $\theta(w) = \sum_t w_t \theta_t$. By Taylor's theorem with integral remainder, for each t,

$$f_{\theta_t}(x) = f_{\theta(w)}(x) + J_x(\theta(w))(\theta_t - \theta(w)) + r_t(x), \tag{134}$$

where J_x is the Jacobian with respect to θ and the remainder satisfies

$$\|r_t(x)\| \le \frac{H_f(x)}{2} \|\theta_t - \theta(w)\|^2 . \tag{135}$$

Average over t with weights w_t:

$$\bar{f}_w(x) = f_{\theta(w)}(x) + J_x(\theta(w)) \sum_t w_t(\theta_t - \theta(w)) + \sum_t w_t r_t(x). \tag{136}$$

Because $\theta(w) = \sum_t w_t \theta_t$,

$$\sum_t w_t(\theta_t - \theta(w)) = 0. \tag{137}$$

Hence

$$\bar{f}_w(x) - f_{\theta(w)}(x) = \sum_t w_t r_t(x), \tag{138}$$

so

$$\left\|\bar{f}_w(x) - f_{\theta(w)}(x)\right\| \leq \sum_t w_t \left\|r_t(x)\right\| \leq \frac{H_f(x)}{2} \sum_t w_t \left\|\theta_t - \theta(w)\right\|^2. \tag{139}$$

Finally, because $\ell(\cdot, y)$ is L_ℓ-Lipschitz,

$$\left|\ell(f_{\theta(w)}(x), y) - \ell(\bar{f}_w(x), y)\right| \leq L_\ell \left\|f_{\theta(w)}(x) - \bar{f}_w(x)\right\| \tag{140}$$

$$\leq \frac{L_\ell H_f(x)}{2} \sum_t w_t \left\|\theta_t - \theta(w)\right\|^2. \tag{141}$$

This is exactly Eq. (46). □

B Additional experimental placeholders

B.1 Implementation details

TBD

B.2 Compute budget

TBD

B.3 Per-dataset tables

TBD

B.4 Sensitivity plots

TBD

B.5 Failure cases

TBD